
CS 223 Digital Design Laboratory Assignment 3

Smart Laundry Service Controller

Preliminary Report Due: April 12, 2026 23:59

Lab Dates and Times

Section 1: April 15, 2026 Wed. 08:30-12:20 in EA-Z04

Section 2: April 13, 2026 Mon. 13:30-17:20 in EA-Z04

Section 3: April 16, 2026 Thur. 08:30-12:20 in EA-Z04

Section 4: April 14, 2026 Tue. 08:30-12:20 in EA-Z04

Location: EA-Z04 (in the EA building, straight ahead past the elevators)

Groups: Each student will do the lab individually. Group size = 1

Important Notes:

- Include your code as plain text, not image, Upload your report in the PDF format.
- You should not upload reports with handwriting. Use computers to write your reports, for circuit schematics, you can use online tools such as Draw.io or Vivado schematics.
- Name your reports in the format: `Name_Surname_ID.pdf`
- Your report should have a cover page including: course code, course name and section, the number of the lab, your name-surname, student ID, date.
- **Mentioned criteria above are mandatory, if any of them are not followed, the preliminary work will not be accepted.**

Preliminary Work [30]

Note: This part must be completed before coming to the lab. Prepare a neatly organized report of your work and submit on Moodle before the start of the lab. Include your code as **plain text, not image!**

Smart Laundry Service Controller

A self-service laundry facility is being automated to improve machine allocation and reduce waiting time. At the control panel, customers select one of four service types. The system assigns the request to one matching machine if available; otherwise the request is placed into a waiting queue for that service type. The maximum queue depth is not specified; choose a reasonable value, state it clearly, and justify your choice in your report. Each machine type has one dedicated machine and a fixed service time:

- Quick Wash: $T_q = 4$ seconds
- Heavy Wash: $T_h = 7$ seconds
- Dryer: $T_d = 9$ seconds
- Delicate Wash: $T_l = 11$ seconds

For simplicity, assume the controller processes service requests sequentially, handling at most one new request per clock cycle. If multiple service buttons are pressed at the same time, requests are queued and processed in a fixed priority order from left to right on the board. When a machine finishes its service, the next waiting request of the same type must start immediately if one exists; otherwise the machine becomes idle.

Design a **finite state machine (FSM)** to manage the machines and the waiting queues under the above constraints.

- [4] Indicate which type of FSM (Moore or Mealy) you will implement and justify your choice. **Note:** Consider factors such as timing behavior, output stability, and implementation complexity when making your decision.
- [10] Provide the state transition diagram, state encoding, state transition and output tables, next-state and output equations, and the FSM schematic.
- [4] Determine the minimum number of flip-flops required by your encoding.
- [4] Redesign your outputs using decoders to simplify the combinational logic. Use a 3-to-8 decoder to generate one-hot encoded state signals from your state register bits, then express your output functions in terms of these decoded signals.
- [8] Write a SystemVerilog module implementing your FSM and a self-checking testbench that verifies functionality.

Smart Laundry Service Controller on FPGA [70]

Implement your design on the Basys3 FPGA board with the following constraints:

1. Use four of the on-board push buttons to submit service requests: Quick Wash, Heavy Wash, Dryer, and Delicate Wash.
2. Display the *current number of busy machines* (0–4) on the 7-segment display in decimal.
3. Use the 16 on-board LEDs to indicate machine status using the pattern `QQQ.HHH.DDD.LLL.`: Solid ON → machine idle, Blinking (1–2 Hz) → machine busy. The dot positions are off.
4. The system must automatically start a queued request when its corresponding machine becomes free.

Clock and Timing:

The Basys3 board provides a 100 MHz system clock. You must implement a clock divider to derive correct 1 Hz timing so that service durations (4 s, 7 s, 9 s, 11 s) are met in real time. Document your clock divider design in your report.

Evaluation:

- [10] Display Test: The 7-segment display must correctly show the current number of busy machines (0–4) in decimal format. The display should update immediately when a service starts or finishes.
- [10] LED Indication Test: Each machine's LEDs must correctly show solid ON when idle and blink at approximately 1–2 Hz when busy. The TA will verify the LED patterns match the specified `QQQ.HHH.DDD.LLL.` arrangement.
- [10] Functionality Test: Button presses must correctly trigger service requests, with proper queueing when a machine is busy. The system should handle simultaneous button presses according to the specified priority rule.
- [40] State Machine Test: Comprehensive evaluation of FSM behavior including: correct state transitions, proper timing adherence (machines become idle after the specified durations), correct queue handling, correct idle behavior when all machines are free, and correct restart when new requests arrive. The instructor will test edge cases such as repeated button presses and mixed request types.

Lab Policies

1. Clean up your lab station, leave your lab workstation for others the way you would like to find it.
2. There are three computers in each row in the lab. Don't use middle computers, unless you are allowed by lab coordinator.
3. You borrow a lab-board containing the development board, connectors, etc. in the beginning. The lab coordinator takes your signature. When you are done, return it to his/her, otherwise you will be responsible and lose points.

4. You must be in the lab, working on the lab, from the time lab starts until you finish and leave. (bathroom and snack breaks are the exception to this rule). Absence from the lab, at any time, is counted as absence from the whole lab that day.
5. No cell phone usage during lab. Tell friends not to call during the lab hours—you are busy learning how digital circuits work!
6. Internet usage is permitted only to lab-related technical sites. No Facebook, Twitter, email, news, video games, etc—you are busy learning how digital circuits work !
7. **If you come to lab later than 30 minutes, you will lose that session completely.**